

```
In [31]: import pandas as pd
```

```
In [32]: sql = pd.read_csv(r'C:\Users\ANIL\Downloads\dataset_1_202511181008.csv')
```

```
In [33]: sql
```

Out[33]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns

```
In [8]: sql[["weather", "temperature"]]
```

Out[8]:

	weather	temperature
0	Sunny	55
1	Sunny	80
2	Sunny	80
3	Sunny	80
4	Sunny	80
...	...	...
12679	Rainy	55
12680	Rainy	55
12681	Snowy	30
12682	Snowy	30
12683	Sunny	80

12684 rows × 2 columns

```
In [9]: sql.head(10)
```

Out[9]:

	destination	passanger	weather	temperature	time	coupon	expiration	gender
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	Female
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	Female
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	Female
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	Female
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	Female
5	No Urgent Place	Friend(s)	Sunny	80	6PM	Restaurant(<20)	2h	Female
6	No Urgent Place	Friend(s)	Sunny	55	2PM	Carry out & Take away	1d	Female
7	No Urgent Place	Kid(s)	Sunny	80	10AM	Restaurant(<20)	2h	Female
8	No Urgent Place	Kid(s)	Sunny	80	10AM	Carry out & Take away	2h	Female
9	No Urgent Place	Kid(s)	Sunny	80	10AM	Bar	1d	Female

10 rows × 27 columns



```
In [10]: sql['passanger'].unique()
```

Out[10]: array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)

```
In [11]: sql[sql['destination']=='Home']
```

Out[11]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
13	Home	Alone	Sunny	55	6PM	Bar	1d	F
14	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	F
15	Home	Alone	Sunny	80	6PM	Coffee House	2h	F
35	Home	Alone	Sunny	55	6PM	Bar	1d	
36	Home	Alone	Sunny	55	6PM	Restaurant(20-50)	1d	
...	...	...	...	...	...	...	...	
12675	Home	Alone	Snowy	30	10PM	Coffee House	2h	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	

3237 rows × 27 columns



In [12]:

```
sql.sort_values('coupon')
```

Out[12]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
11702	Home	Partner	Sunny	30	10PM	Bar	2h	F
9930	No Urgent Place	Alone	Snowy	30	2PM	Bar	1d	F
10632	Home	Alone	Rainy	55	6PM	Bar	1d	
7997	No Urgent Place	Friend(s)	Rainy	55	10PM	Bar	2h	
11166	Work	Alone	Snowy	30	7AM	Bar	1d	F
...	...	...	...	...	...	...	...	
10476	Home	Alone	Sunny	80	6PM	Restaurant(<20)	1d	F
5447	Home	Alone	Sunny	80	10PM	Restaurant(<20)	2h	F
10478	Home	Alone	Snowy	30	10PM	Restaurant(<20)	2h	F
5440	No Urgent Place	Alone	Sunny	80	2PM	Restaurant(<20)	2h	F
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F

12684 rows × 27 columns



In [13]:

```
sql.rename(columns={'destination':'Destination'},inplace=True)
```

In [14]:

```
sql
```

Out[14]:

	Destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	I
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	I
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	I
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	I
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	I
...	...	...	...	...	...	...	...	
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns



In [15]:

```
sql.groupby('occupation').size().to_frame('Count').reset_index()
```

Out[15]:

	occupation	Count
0	Architecture & Engineering	175
1	Arts Design Entertainment Sports & Media	629
2	Building & Grounds Cleaning & Maintenance	44
3	Business & Financial	544
4	Community & Social Services	241
5	Computer & Mathematical	1408
6	Construction & Extraction	154
7	Education&Training&Library	943
8	Farming Fishing & Forestry	43
9	Food Preparation & Serving Related	298
10	Healthcare Practitioners & Technical	244
11	Healthcare Support	242
12	Installation Maintenance & Repair	133
13	Legal	219
14	Life Physical Social Science	170
15	Management	838
16	Office & Administrative Support	639
17	Personal Care & Service	175
18	Production Occupations	110
19	Protective Service	175
20	Retired	495
21	Sales & Related	1093
22	Student	1584
23	Transportation & Material Moving	218
24	Unemployed	1870

In [16]: `sql.groupby('weather')['temperature'].mean().to_frame('avg_temp').reset_index()`

```
Out[16]:
```

	weather	avg_temp
0	Rainy	55.000000
1	Snowy	30.000000
2	Sunny	68.946271

```
In [17]: sql.groupby('weather')['temperature'].nunique().to_frame('count_distinct_temp').res
```

```
Out[17]:
```

	weather	count_distinct_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

```
In [19]: sql.groupby('weather')['temperature'].sum().to_frame('sum_temp').reset_index()
```

```
Out[19]:
```

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

```
In [30]: sql.groupby('weather')['temperature'].min().to_frame('min_temp').reset_index()
```

```
Out[30]:
```

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

```
In [21]: sql.groupby('weather')['temperature'].max().to_frame('max_temp').reset_index()
```

```
Out[21]:
```

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

```
In [22]: sql.groupby('occupation').filter(lambda x:x['occupation'].iloc[0]=='Student').group
```

```
Out[22]: occupation
Student    1584
dtype: int64
```



```
In [44]: sql1 = pd.read_csv(r'C:\Users\ANIL\Downloads\dataset_1_202511181008.csv')
```

```
In [45]: sql1
```

```
Out[45]:
```

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM	Carry out & Take away	1d	
12680	Work	Alone	Rainy	55	7AM	Carry out & Take away	1d	
12681	Work	Alone	Snowy	30	7AM	Coffee House	1d	
12682	Work	Alone	Snowy	30	7AM	Bar	1d	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

12684 rows × 27 columns



```
In [51]: pd.concat([sql,sql1])['destination'].drop_duplicates()
```

```
Out[51]: 0    No Urgent Place
        13           Home
        16           Work
        Name: destination, dtype: object
```

```
In [52]: pd.merge(sql, sql1[['time', 'part_of_day']], on='time', how='inner')[['destination']
```

```

-----
KeyError                                Traceback (most recent call last)
Cell In[52], line 1
----> 1 pd.merge(sql, sql1[['time', 'part_of_day']], on='time', how='inner')[['destination', 'time', 'part_of_day']]

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\frame.py:4108, in DataFrame.__getitem__(self, key)
    4106     if is_iterator(key):
    4107         key = list(key)
-> 4108     indexer = self.columns._get_indexer_strict(key, "columns")[1]
    4110 # take() does not accept boolean indexers
    4111 if getattr(indexer, "dtype", None) == bool:

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6200, in Index._get_indexer_strict(self, key, axis_name)
    6197 else:
    6198     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 6200 self._raise_if_missing(keyarr, indexer, axis_name)
    6202 keyarr = self.take(indexer)
    6203 if isinstance(key, Index):
    6204     # GH 42790 - Preserve name from an Index

File C:\ProgramData\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6252, in Index._raise_if_missing(self, key, indexer, axis_name)
    6249     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
    6251 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
-> 6252 raise KeyError(f"{not_found} not in index")

KeyError: "['part_of_day'] not in index"

```

In []:

In []:

In [53]: `sql[sql1['weather'].str.startswith('Sun')]`

Out[53]:

	destination	passanger	weather	temperature	time	coupon	expiration	g
0	No Urgent Place	Alone	Sunny	55	2PM	Restaurant(<20)	1d	F
1	No Urgent Place	Friend(s)	Sunny	80	10AM	Coffee House	2h	F
2	No Urgent Place	Friend(s)	Sunny	80	10AM	Carry out & Take away	2h	F
3	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	2h	F
4	No Urgent Place	Friend(s)	Sunny	80	2PM	Coffee House	1d	F
...	...	...	...	...	...	...	...	
12673	Home	Alone	Sunny	30	6PM	Carry out & Take away	1d	
12676	Home	Alone	Sunny	80	6PM	Restaurant(20-50)	1d	
12677	Home	Partner	Sunny	30	6PM	Restaurant(<20)	1d	
12678	Home	Partner	Sunny	30	10PM	Restaurant(<20)	2h	
12683	Work	Alone	Sunny	80	7AM	Restaurant(20-50)	2h	

10069 rows × 27 columns



In [54]:

```
sql[(sql['temperature'] >= 29) & (sql['temperature'] <= 75)]['temperature'].unique()
```

Out[54]: array([55, 30], dtype=int64)

In [55]:

```
sql[sql['occupation'].isin(['Sales & Related', 'Management'])]['occupation']
```

Out[55]:

occupation	
193	Sales & Related
194	Sales & Related
195	Sales & Related
196	Sales & Related
197	Sales & Related
...	...
12679	Sales & Related
12680	Sales & Related
12681	Sales & Related
12682	Sales & Related
12683	Sales & Related

1931 rows × 1 columns

In []: